

# Generation Interconnection System Impact Study Report

## For Queue Project Q14-0389

Deer Park Energy Center 100 kV

January 20, 2016

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0389. The Interconnection Customer proposes a 115.2 MW battery energy storage facility interconnecting at the Deer Park Energy Center 100 kV substation in the Illinois area. The request was received on October 08, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,081,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DEER PARK ENERGY CTR-COMED 100 KV Ckt 1  
Contingency: COMED-DEER PARK ENERGY CTR 100 KV Line 2  
Loading: 154.0% (Rating: 143 MVA)  
DFAX: 83.5%  
Pre-project loading: 86.8%
2. Facility: DEER PARK ENERGY CTR-COMED 100 KV #2  
Contingency: COMED-DEER PARK ENERGY CENTER 100 KV  
Loading: 154.0% (Rating: 143 MVA)  
DFAX: 83.5%  
Pre-project loading: 86.8%
3. Facility: RICHLAND-TWIN COUNTY RANCH 230 KV  
Contingency: SWAN VLY-TREGO 500 KV  
Loading: 104.1% (Rating: 543 MVA)  
DFAX: 8.0%  
Pre-project loading: 102.4%

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                       | Required Upgrade                               | Cost Estimate | Allocation to Project |
|------------------------------------------------|------------------------------------------------|---------------|-----------------------|
| DEER PARK ENERGY CTR-COMED<br>100 KV Circuit 1 | Rebuild line with higher-capacity<br>conductor | \$499,000     | 100% (\$499,000)      |

| Facility                                   | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------------------|---------------------------------------------|---------------|-----------------------|
| COMED-DEER PARK ENERGY CENTER 100 KV Ckt 2 | Rebuild line with higher-capacity conductor | \$441,000     | 100% (\$441,000)      |
| TWIN COUNTY RANCH-RICHLAD 230 KV L         | Reconductor line                            | \$12,535,000  | 1% (\$141,000)        |
|                                            | Total allocated to project                  |               | \$1,081,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.